Matemática Discreta 2

Aula 1 - Apresentação da Disciplina

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Apresentação

CIn - UFPE

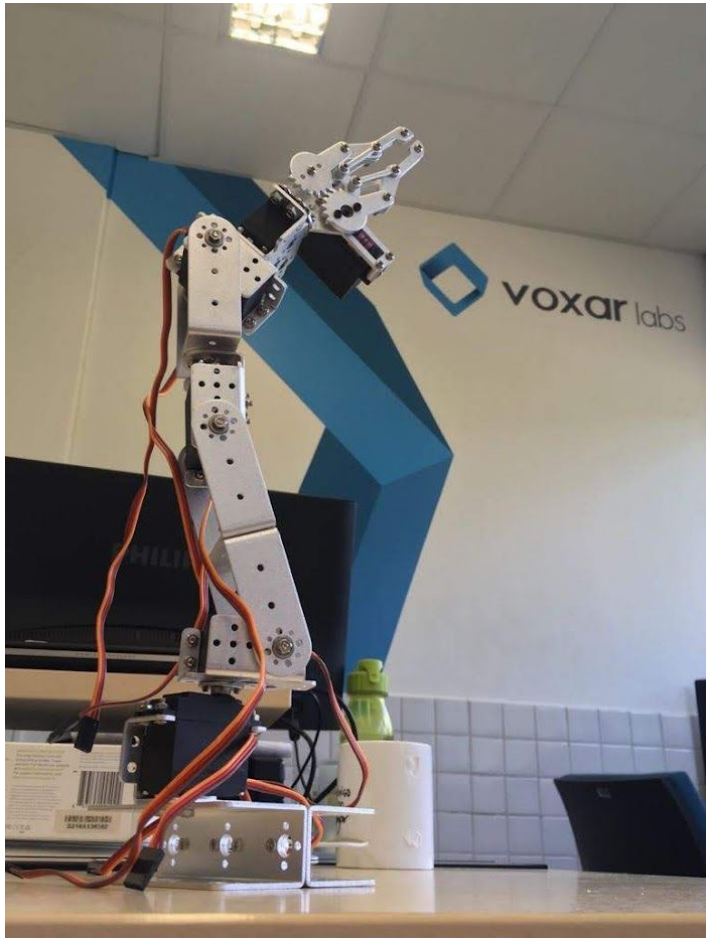
Voxar Labs

Co-fundador (2011)

Parcerias com a indústria

- LG
- Samsung
- HP
- Marinha

Líder de pesquisas



Uninassau

Disciplinas

- Computação Gráfica
- Inteligência Artificial
- ...

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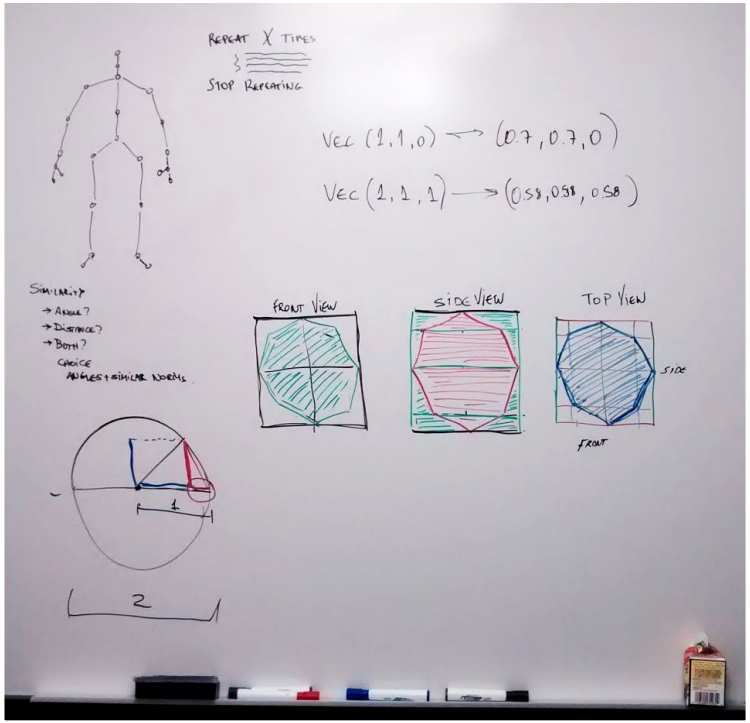
Microsoft Research



Projetos Anteriores

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Trabalho de graduação: AirGuitarFramework

SBGames 2009

An open-source **framework** for **air guitar** games

LS Figueiredo, JMXN Teixeira… - 2009 VIII Brazilian …, 2009 - ieeexplore.ieee.org

This paper presents an open-source **framework** for developing **guitar**-based games using gesture interaction. The goal of this work was to develop a robust platform capable of providing seamless real time interaction, intuitive playability and coherent sound output …

☆ 99 Cited by 17 Related articles All 15 versions

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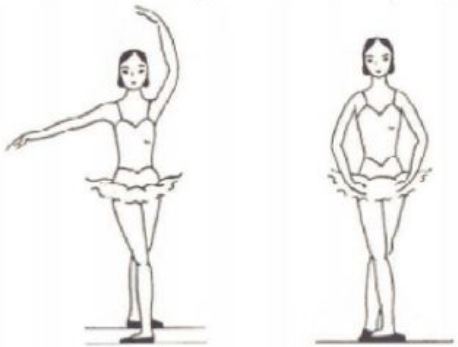
Linguagem de Domínio Específico (DSL) para Interações Gestuais

Declarations

app	::=	APP id : (*gesture* .) + EOF
gesture	::=	GESTURE id : *pose* + *execution*
pose	::=	POSE id : *statement* (, *statement*) * .
statement	::=	*transform* | *restriction*
execution	::=	EXECUTION : (*repeat* the following steps number *executionStep*(, *executionStep*) * | *executionStep*(, *executionStep*) *)
executionStep	::=	*motionConstraint* ? id (and *holdConstraint*) ?

Transforms

transform	::=	*pointTo* | *rotatePlane* | *rotateDirection*
pointTo	::=	point your ? *bodyPart*((, your ? *bodyPart*) * and your ? *bodyPart*) ? (to | to your) ? *direction*
rotatePlane	::=	rotate your *bodyPart*((, your ? *bodyPart*) * and your ? *bodyPart*) ? number degrees *angularDirection* on the ? *referencePlane*
rotateDirection	::=	rotate your *bodyPart* ((, your ? *bodyPart*) * and your ? *bodyPart*) ? number degrees (to | to your) ? *direction*



(b) Ballet poses.

```
GESTURE fourth-position-en-avant:
    POSE cross-legs-one-behind-the-other:
        put your left ankle behind your right ankle,
        put your left ankle to the right
                        of your right ankle.
        // do not connect your ankles.

    POSE high-arc-arms-to-right:
        point your arms down,
        rotate your right arm 70 degrees up,
        rotate your left elbow 20 degrees to your left,
        rotate your left wrist 25 degrees to your right.

    EXECUTION:
        // fourth-position-en-avant-composed
        stand-straight,
        point-feet-out,
        stretch-legs,
        cross-legs-one-behind-the-other,
        high-arc-arms-to-right.
```

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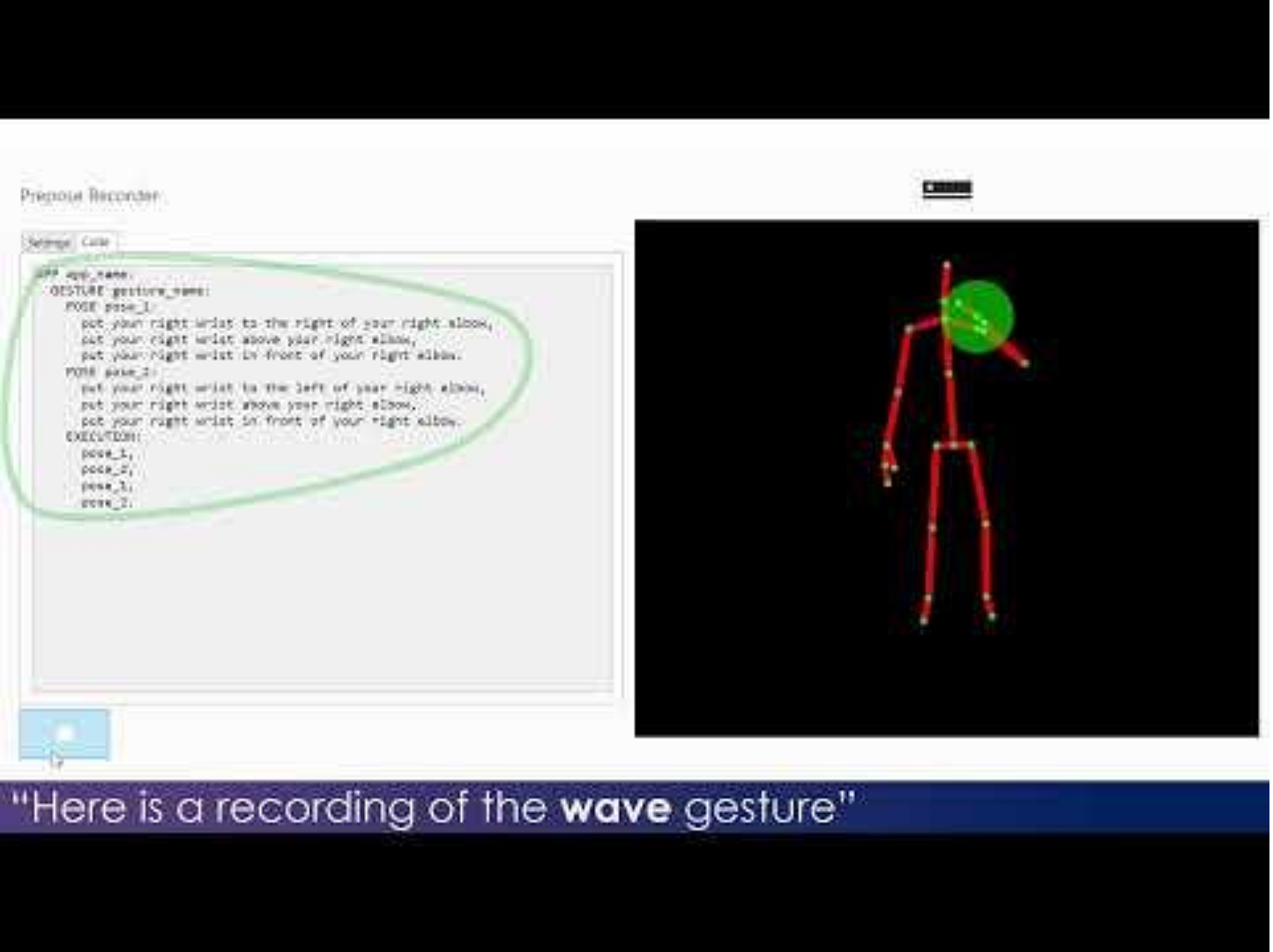

Linguagem de Domínio Específico (DSL) para Interações Gestuais

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Reconhecimento de Gestos em Tempo Real

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Funcionamento da Disciplina

Funcionamento

Comunicação:

- Grupo no aplicativo Telegram
- Contatos diretos em casos de urgência

Aulas e Materiais:

- Google Classroom

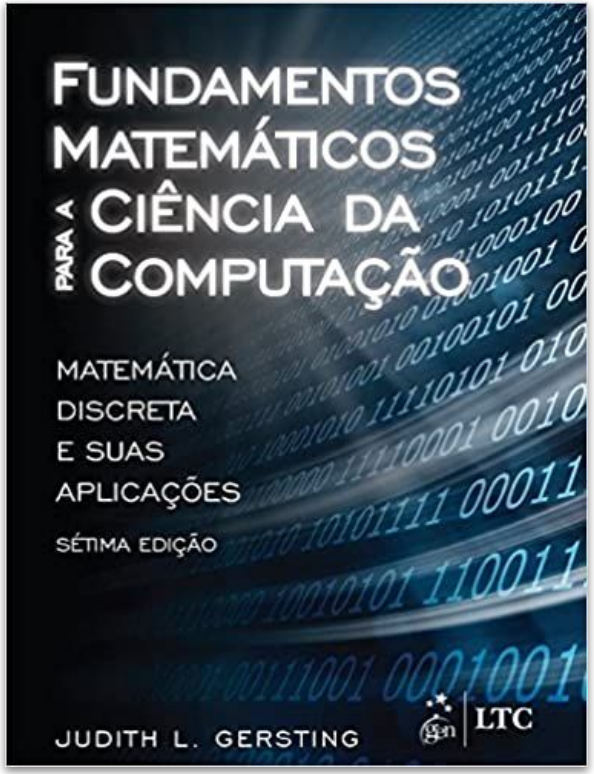
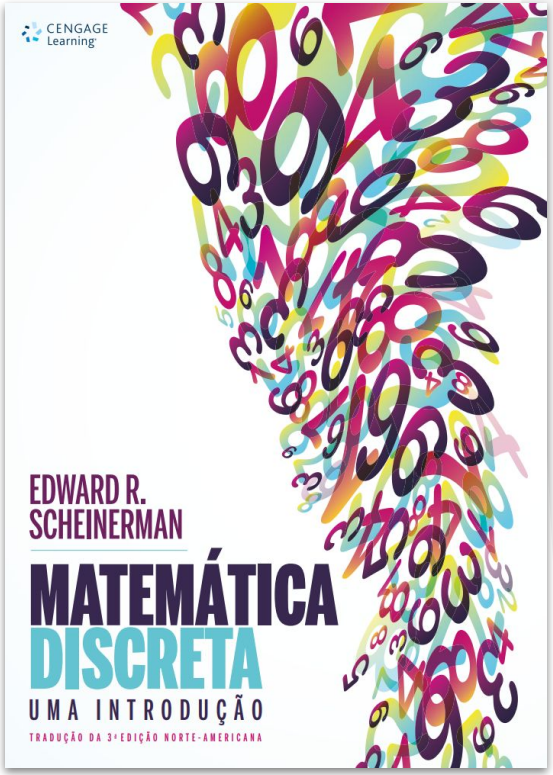
Formato das Notas:

- 50% Mini-provas
- 50% Provas

Referências

Livro:

- Fundamentos Matemáticos para Ciência da Computação (7a Edição)
- Matemática Discreta - Uma Introdução (3a Edição)

Ferramentas:

- Qualquer IDE de programação para atividades práticas extras